

# Matthew Frazier

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AI Researcher and Engineer with 10+ years of experience building data, analytics, and distributed systems, with a focus on designing and evaluating LLM-based applications. Proven track record prototyping AI systems and validating their pedagogical efficacy through empirical, mixed-methods research. Applied prompt engineering, retrieval-augmented generation (RAG), and conversational system design to create AI-driven learning environments. Ph.D. in Computer and Information Sciences with research in conversational agents intersecting HCI and applied AI.

## CORE SKILLS

**AI & Systems:** LLMs, RAG, prompt engineering (CoT, ReAct, few-shot), human-in-the-loop (HITL), vector search, inference pipelines, LangChain, LangGraph, CrewAI, Qdrant, Langfuse, MLflow, LLM-as-a-judge

**Programming & Engineering:** Python, SQL, TypeScript/JavaScript, microservices, REST APIs, system design

**Data & Infrastructure:** Airflow, Argo Workflows, AWS (Lambda, S3, SQS), Redshift, Snowflake, distributed data pipelines

**Research Methods:** Mixed-methods research, empirical evaluation, survey design (Qualtrics), qualitative analysis, experimental design, usability testing, concept testing, participatory design, user interviews

## RESEARCH & TEACHING EXPERIENCE

**PhD Researcher & Instructor** | University of Delaware | Newark, DE 2018 - 2024

- Led research on conversational agents for computer science education, designing and developing LLM-based prototype systems using prompt engineering techniques (e.g., RAG, CoT, few-shot) through mixed-methods empirical studies, resulting in peer-reviewed publications. Mentored undergraduate researchers.
- Engineered and iterated on conversational AI system behavior and interaction patterns to guide student reasoning through structured Socratic dialogue rather than direct answer generation.
- Instructed undergraduate computer science courses. Created curriculum and taught programming, data structures, and computer science ethics.

**Graduate Assistant & Researcher** | West Chester University of Pennsylvania | West Chester, PA 2016 - 2017

- Evaluated data structures assignments (Java), assessing correctness, efficiency, and algorithmic approach.
- Developed an object detection and identity matching application (computer vision - Python, OpenCV) that analyzes university parking lot videos to detect open parking spots for student lot availability.

## PROFESSIONAL EXPERIENCE (select)

**Senior Software Engineer** (Data Science, Scaled Media Solutions) | Goodway Group | New York, N.Y. 2021 - Present

- Designed and implemented AI-driven analytics systems within the **GOES™** (Goodway Outcome Engineering System) platform, developing LLM-based systems for natural language querying and hierarchical insight generation. Built production-grade AI workflows integrating vector search, RAG, stateful memory, prompt engineering (CoT, ReAct, few-shot), and streaming inference via REST APIs to decision-support systems.
- Improved production stability of **RealValue™**, a distributed optimization system processing **40M+** impression-level decisions daily across campaign datasets, built on microservices, ETL pipelines, and DSP integrations (e.g., The Trade Desk). Re-architected error handling (Rollbar, Datadog) and validation logic within processing systems, increasing system reliability while reducing production errors by **90%**.
- Integrated backend microservices and REST APIs across **5+** external platforms (The Trade Desk, Bing, YouGov, Google Ads, Meta). Built high-volume ETL data pipelines (Airflow, AWS MWAA, Argo Workflows, Redshift, Snowflake, S3) supporting large-scale data and analytics systems. Led DevOps initiatives and co-led AI initiatives with Data Science leadership, driving best practices and cross-team MLOps governance alignment across platform and AI systems.

- Architected full-stack and event-driven distributed systems, including AWS Lambda, SNS, SQS (FIFO), and S3 workflows to automate YouGov analysis workflows, replacing manual Jupyter notebook execution with asynchronous processing. Deployed frontend (React/Next.js, Streamlit) and data visualization interfaces for real-time insights.

**Senior Technical Operations Engineer** (Business Intelligence) | Goodway Group | New York, N.Y. 2018 - 2021

- Owned data reliability, integrity, and availability of marketing analytics systems. Managed end-to-end data operations and reporting infrastructure (Airflow, Redshift, S3, Tableau) under strict daily data freshness SLAs.
- Owned org-wide incident response, root-cause analysis, postmortem, and data discrepancy triage across ETL pipeline and reporting systems. Adopted preventative fixes and monitoring workflows that improved system stability and reduced recurring issues.

**Principal Campaign Operations Engineer** | Goodway Group | New York, N.Y. 2014 - 2018

- Launched **COLT** (Cacidi On-Line Tool), a web-based automation system augmenting Adobe InDesign templates that parameterized ad copy and legal logic. Automated campaign production, reducing creative approval time by approximately **80% (from 10-12 business days to 2 business days)** and enabling scalable execution across multi-location automotive clients (e.g., Chevrolet).
- Executed, QA'd, and reported on programmatic campaigns across platforms (The Trade Desk, Google Ads, Meta), collaborating with ad operations and media trading to improve data-driven campaign pacing and performance.

**Senior Search Engine Optimization (SEO) Strategist** | IONOS (1&1 Internet) | Chesterbrook, PA 2012 - 2014

- Established and analyzed SEO and Search Engine Marketing (SEM) strategies through copy and creative variation A/B testing, metatag optimization, user engagement behavior analysis (CTR, time-on-page), and keyword analysis, improving organic visibility and traffic across **300+** small business websites in US, UK, and Canadian markets (Google Analytics, SEMrush).

## EDUCATION

**University of Delaware** | Newark, DE

Ph.D., Computer and Information Sciences (Magna Cum Laude) 2024

**West Chester University of Pennsylvania** | West Chester, PA

M.Sc., Computer Science (Cum Laude) 2017

B.Sc., Marketing 2010

## PUBLICATIONS (select)

- Customizing ChatGPT to Help Computer Science Principles Students Learn Through Conversation (ITiCSE 2024)
- Investigating User Perceptions of Conversational Agents for Software-related Exploratory Web Search (ICSE 2022)